

Department Overview

- **Department Name:** Department of Chemistry
- **Faculty:** Faculty of Basic Sciences
- **History:** Began regular operations in 2015, launching both its undergraduate and postgraduate frameworks.
- **Standards & Mode:** The curriculum aligns strictly with the Higher Education Commission (HEC) of Pakistan guidelines. Programs are offered in both **morning and evening** sessions.
- **Faculty Profile:** Features a highly qualified academic team including **11 PhD degree holders** with vast national and international teaching and research exposure.
- **Core Disciplines of Specialization:**
 1. Analytical Chemistry
 2. Biochemistry
 3. Inorganic Chemistry
 4. Organic Chemistry
 5. Physical Chemistry

Mission & Objectives

- **Mission:** To create a conducive environment of learning for its undergraduate and postgraduate students and to achieve excellence in the area of research and education.
- **Objective:** To educate the next generation of chemists with the capacity to solve real problems and provide leadership that can lead to the economic transformation of society.

1. Bachelor of Science in Chemistry (BS Chemistry)

Admission Requirements

- **Academic Background:** F.Sc with Chemistry from any recognized board (or equivalent qualification) with at least **45% marks**.
- **Testing:** Successfully qualifying an aptitude test with a percentage determined by the Admission Committee.

Degree Requirements

- **Total Credit Hours:** 140
- **Total Courses:** 44
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Foundation & Core Curriculum (Semesters 1 to 6)

Note: Credit hours are listed as (Theory + Lab).

- **1st Semester:** Inorganic Chemistry-I (3+1), Introduction to Computer (2+1), Mathematics-I (3+0), Pak. Studies (2+0), Functional English (3+0), Physics (2+1). **Total: 15+3 = 18 CH**

- **2nd Semester:** Organic Chemistry-I (3+1), Environmental Chemistry (3+0), Food and Nutrition Chemistry (3+0), Mathematics-II (3+0), Islamic Studies / Ethics (2+0), Communication Skills (3+0). **Total: 17+1 = 18 CH**
- **3rd Semester:** Physical Chemistry-I (3+1), Basic Forensic Chemistry (3+0), Industrial Chemistry (3+0), Introduction to Psychology (2+0), Statistics (3+1), Academic Reading & Writing (3+0). **Total: 17+2 = 19 CH**
- **4th Semester:** Analytical Chemistry-I (3+1), Applied Chemistry (3+0), Biochemistry-I (3+1), Soil Chemistry (3+0), Research Methods (3+0), Fuel Chemistry (3+0). **Total: 18+2 = 20 CH**
- **5th Semester:** Biochemistry-II (3+1), Inorganic Chemistry-II (3+1), Organic Chemistry-II (3+1), Physical Chemistry-II (3+1). **Total: 12+4 = 16 CH**
- **6th Semester:** Analytical Chemistry-II (3+1), Inorganic Chemistry-III (3+1), Organic Chemistry-III (3+1), Physical Chemistry-III (3+1). **Total: 12+4 = 16 CH**

Specialization Pathways (Semesters 7 & 8)

Students select one of the five specific specialization branches during their final year. Each track mandates **12 Theory CH** and **4 Lab CH** per semester.

Track A: Inorganic Chemistry

- **7th Semester:** Inorganic Reaction Mechanism, π -Acceptor Ligands and Inorganic Polymer (3+0); Recent Research in Inorganic Chemistry (3+0); Inorganic Spectroscopy (3+0); Bio-Inorganic Chemistry (3+0); Inorganic Special Lab-I (0+1); Research Thesis / Project / Position Paper (0+3).
- **8th Semester:** Organometallics (3+0); Symmetry and Magneto-chemistry (3+0); Radio and Nuclear Chemistry (3+0); Nanochemistry (3+0); Inorganic Special Lab-II (0+1); Research Thesis / Project / Position Paper (0+3).

Track B: Organic Chemistry

- **7th Semester:** Heterocyclic and Organometallic Compounds (3+0); Reactive Intermediates (3+0); Recent Research in Organic Chemistry (3+0); Organic Spectroscopy (3+0); Organic Special Lab-I (0+1); Research Thesis / Project / Position Paper (0+3).
- **8th Semester:** Natural Products (3+0); Organic Synthesis (3+0); Nanochemistry (3+0); Medicinal Chemistry (3+0); Organic Special Lab-II (0+1); Research Thesis / Project / Position Paper (0+3).

Track C: Physical Chemistry

- **7th Semester:** Electrochemistry and Statistical Thermodynamics (3+0); Recent Research in Physical Chemistry (3+0); Quantum Chemistry (3+0); Molecular Spectroscopy (3+0); Physical Special Lab-I (0+1); Research Thesis / Project / Position Paper (0+3).
- **8th Semester:** Chemical Kinetics (3+0); Nanochemistry (3+0); Colloid and Surface

Chemistry (3+0); Polymer Chemistry (3+0); Physical Special Lab-II (0+1); Research Thesis / Project / Position Paper (0+3).

Track D: Biochemistry

- **7th Semester:** Metabolism and Immunology (3+0); Protein Chemistry and Enzymology (3+0); Recent Research in Biochemistry (3+0); Molecular Biology (3+0); Biochemistry Special Lab-I (0+1); Research Thesis / Project / Position Paper (0+3).
- **8th Semester:** Nutrition and Plant Biochemistry (3+0); Clinical Biochemistry (3+0); Nanochemistry (3+0); Molecular Spectroscopy (3+0); Biochemistry Special Lab-II (0+1); Research Thesis / Project / Position Paper (0+3).

Track E: Analytical Chemistry

- **7th Semester:** Electro-Analytical Techniques (3+0); Advanced Separation Techniques (3+0); Advance Microscopy (3+0); Recent Research in Analytical Chemistry (3+0); Analytical Special Lab-I (0+1); Research Thesis / Project / Position Paper (0+3).
- **8th Semester:** Luminescence Spectroscopy and Thermal Analysis (3+0); Nanochemistry (3+0); Food and Drug Analysis (3+0); Environmental Analysis (3+0); Analytical Special Lab-II (0+1); Research Thesis / Project / Position Paper (0+3).

2. Master of Science in Chemistry (MS Chemistry)

Admission Requirements

- **Academic Background:** 16 years of education in Chemistry or equivalent [e.g., 4-year BS or MSc] from a recognized university with at least **50% marks** or a minimum CGPA of **2.0 out of 4.0**.
- **Standardized Testing:** Fulfilling one of the following testing criteria:
 - GAT-General with a minimum 50% score.
 - GAT-Subjective with a minimum 60% score.
 - GRE Subject Test (International) with at least a 50th percentile rank.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses Required:** 24 (*Note: This text entry on the page typically references total credit hour distributions, as standard HEC MS tracks traditionally feature 8–10 courses*).
- **Minimum Academic Standing:** CGPA $\geq$ 2.5

3. Doctor of Philosophy in Chemistry (PhD Chemistry)

Admission Requirements

- **Academic Background:** An HEC-recognized MS or equivalent 18-year degree in Chemistry. The completed degree must have included a minimum of **30 credit hours** (24 credit hours of graduate level courses + 6 credit hours of thesis) passed with a 1st

Division grading or a minimum CGPA of **3.0 out of 4.0**.

- **Standardized Testing:** Prior to admission confirmation, candidates must pass either the **GRE International Subject Test** with a minimum 60th percentile rank, or a valid **GAT Subject Test** with at least **60% marks**.

Degree Requirements

- **Total Credit Hours:** 18
- **Total Courses Required:** 6 advanced doctoral courses.
- **Minimum Academic Standing:** CGPA $\geq$ 3.0